

Skills Summary

Slope: Upside Down or Sign Flipped?

Use this reference while completing your worksheet or when studying.

The rule, and the two ways it goes wrong:

$$m = \frac{\text{change in } y}{\text{change in } x} = \frac{y_2 - y_1}{x_2 - x_1}$$

Upside down means the x change ended up on top — the answer is the reciprocal of the true slope. **Sign flipped** means the two subtractions started at different points — the digits are right and the sign is wrong.

Skill 1 — Which difference goes on top

Rule: the change in y (rise) always goes on top; the change in x (run) always goes underneath. Check with units: a slope of dollars per hour must have dollars on top.

Example: Find the slope through $(1, 2)$ and $(5, 14)$.

Step 1. The question asks how fast y grows per unit of x , so the y change belongs on top — write the fraction before dividing anything.

Step 2. Change in y is $14 - 2 = 12$; change in x is $5 - 1 = 4$; so the fraction is $12/4$.

The slope is 12 over 4. $m = 3$

Try it: Find the slope through $(2, 5)$ and $(6, 7)$.

$\frac{2}{4} = m$:check

(!) Watch out: if your slope looks like a small fraction where you expected a big number (or the other way round), suspect an upside-down ratio before you suspect the arithmetic.

Skill 2 — Keeping the sign right

Rule: pick a starting point and subtract *that same point* in both differences. Reversing only one of them flips the sign of the answer. A line that falls from left to right must end up with a negative slope.

Example: Find the slope through $(2, 9)$ and $(6, 1)$.

Step 1. The y value drops while x grows, so the answer must come out negative — that is the check to make at the end.

Step 2. Starting from the first point both times: y change $1 - 9 = -8$, x change $6 - 2 = 4$, giving $-8/4$.

Negative, as predicted. $m = -2$

Try it: Find the slope through $(0, 7)$ and $(4, 3)$.

$1 - 7 = m$:check

Skill 3 — Rate of change from a table

Rule: pick any two rows, subtract the outputs for the top of the fraction and the matching inputs for the bottom, keeping the same row first in both. The rate is the same for any pair of rows on a linear table.

Example: Find the rate of change for this table.

x	0	2	4	6
y	5	11	17	23

Step 1. The rate wanted is the output change per input change, so use the first and last rows and put the y change on top.

Step 2. y change $23 - 5 = 18$; x change $6 - 0 = 6$; so the rate is $18/6$.

Six units of x buy 18 units of y . **$m = 3$**

Try it: A table has $x = 1, 5$ with $y = 20, 8$. Find the rate of change.

check: $m = -3$